SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT162

Timed Online Assignment - July Semester 2022

Object Oriented Programming

Thursday, 24 November 2022 04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **FOURTEEN (14)** pages (including cover page).

2. You must answer **FOUR (4)** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

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Answer all questions. (Total 100 marks)

Question 1

Study the program given in Figure Q1 and answer the questions that follows:

```
class MovieCard:
    _TICKET_COUNT = {70:10, 100:15}

    def __init__(self, price):
        # assuming the price must either be 70 and 100
        self._price = price
        self._tickets = MovieCard._TICKET_COUNT[price]

    def tickets(self, newTickets):
        self._tickets = newTickets

    @property
    def tickets(self):
        return self._tickets

    def __str__(self):
        return "Ticket value: ${:.2f} \t Tickets remaining: {}".format
                                                (self._price, self._tickets)

class HSBCMovieCard(MovieCard):
    _TICKET_COUNT = {70:12, 100:18}

    def __init__(self, name, price):
        super().__init__(price)
        self._name = name

    def __str__(self):
        """ string representation of the object """
        return "CardHolder Name: {}\t".format(self._name) +
super().__str__()
```

Figure Q1

Question 1a

Describe and explain what the following statements are doing and what is the output.

```
card1 = MovieCard(70)
print( card1.tickets )
```

(2 marks)

Question 1b

The statements below will execute successfully, and output is shown.

```
card2 = HSBCMovieCard("Jeffery Tan", 100)
print( card2.tickets )

output → 15
```

However, the output is not correct as $100 HSBC MovieCard should entitle to 18 tickets.

- Indicate which statement number(s) in Figure Q1 causes this wrong ticket entitlement.
- Explain why the error occurs and make correction(s) to the identified statement(s) to compute the correct ticket entitlement.

(6 marks)

Question 1c

There is a runtime error when executing the following statements below.

```
card1.tickets = 11
```

Explain why this runtime error occurs and make correction(s) to the part if it is possible to correct the error.

(4 marks)

Question 1d

Due to demand, there is a need to offer 5 tickets at $35 for MovieCard and 6 tickets at $35 for HSBCMovieCard.

Modify the program in Figure Q1, by adding appropriate method(s) such that the codes can easily cater to adding new offering or removing offerings.

(8 marks)

Question 2

Demonstrate your understanding of how class hierarchy and association are used to organize information, by constructing the class hierarchy and association according to specification.

The class diagram in Figure Q2 shows specifications of the food items sold in a school café.

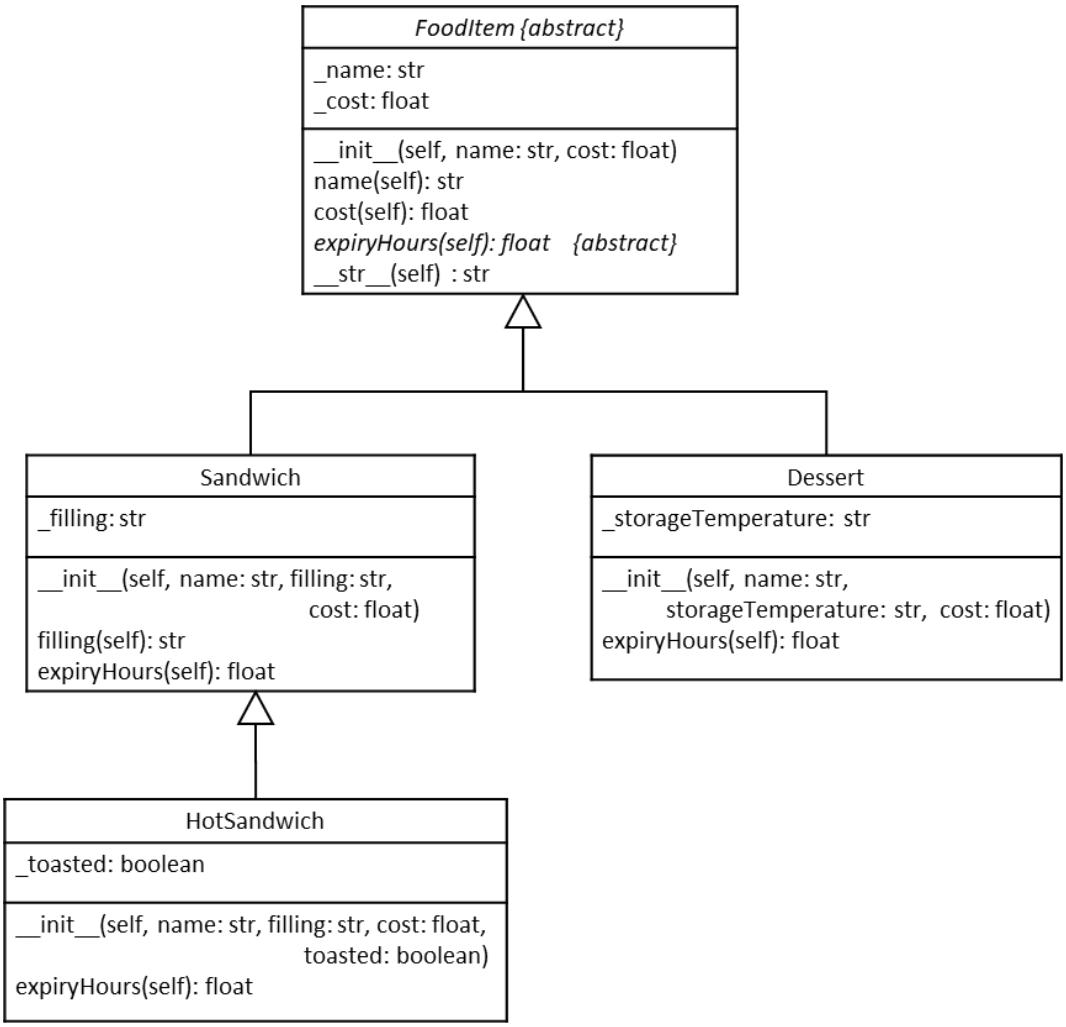

Figure Q2

Question 2a

Implement the abstract class - FoodItem.

The FoodItem class represents a food item that the school café is offering.

- It has the 2 instance variables:
  - _name (str): a description or name for the food item.
  - _cost (float): the cost of the food item.

- Constructor initialises these 2 instance variables: _name and _cost.
- Getter methods for name and cost. Use the property decorator.
- One abstract method, expiryHours which returns the expiry hours of the food item.
- The __str__ method returns a string representation of a FoodItem object, including the name, cost and expiry hours of the FoodItem objects.

This is an example in the suggested format:

```
Name: Strawberry Pudding   Price: $2.00   Expiring in 2 hrs
```

(8 marks)

Question 2b

Write Python class definitions for Sandwich and HotSandwich class.

The Sandwich class is a subclass of FoodItem, and it has:

- One additional instance variable: _fillings (str) describing the main ingredients (such as turkey, ham, cheese etc.) of the sandwich.
- Implement the method expiryHours that returns the standard expiry hours of a sandwich as 2 (hours).

The HotSandwich is a subclass of Sandwich and has additional instance variable _toasted (boolean) that indicate if the bread is toasted.

- The method expiryHours returns expiry hours as 1.5 if this hot sandwich is toasted.

(8 marks)

Question 2c

Explain and describe what type of overriding you have implemented for expiryHours and __str__ methods in class HotSandwich. What is the OO concept that allows for overriding?

(4 marks)

Question 2d

Write Python statements to invoke the methods you have defined.

Question 2d(i)
Create the appropriate FoodItem objects using the information in Table Q2(d) below:

Name	Cost	Toasted
Toasted Cheese Sandwich	$3.90	True
Hot Pastrami Sandwich	$4.80	False
Cold Cut Sandwich	$4.50	-

Table Q2(d)

(3 marks)

Question 2d(ii)
Print the expiry hours for "Toasted Cheese Sandwich". Based on your understanding, what should be the output?

(2 marks)

Question 3

Apply the principles of object-oriented programming and develop a program to help student create their own customised "Snack Box", by adding FoodItem into snack box or removing FoodItem from it. Your program must handle SnackBoxException which you will define for this question. Study the class diagram in Figure Q3(a), which now feature the SnackBox class.

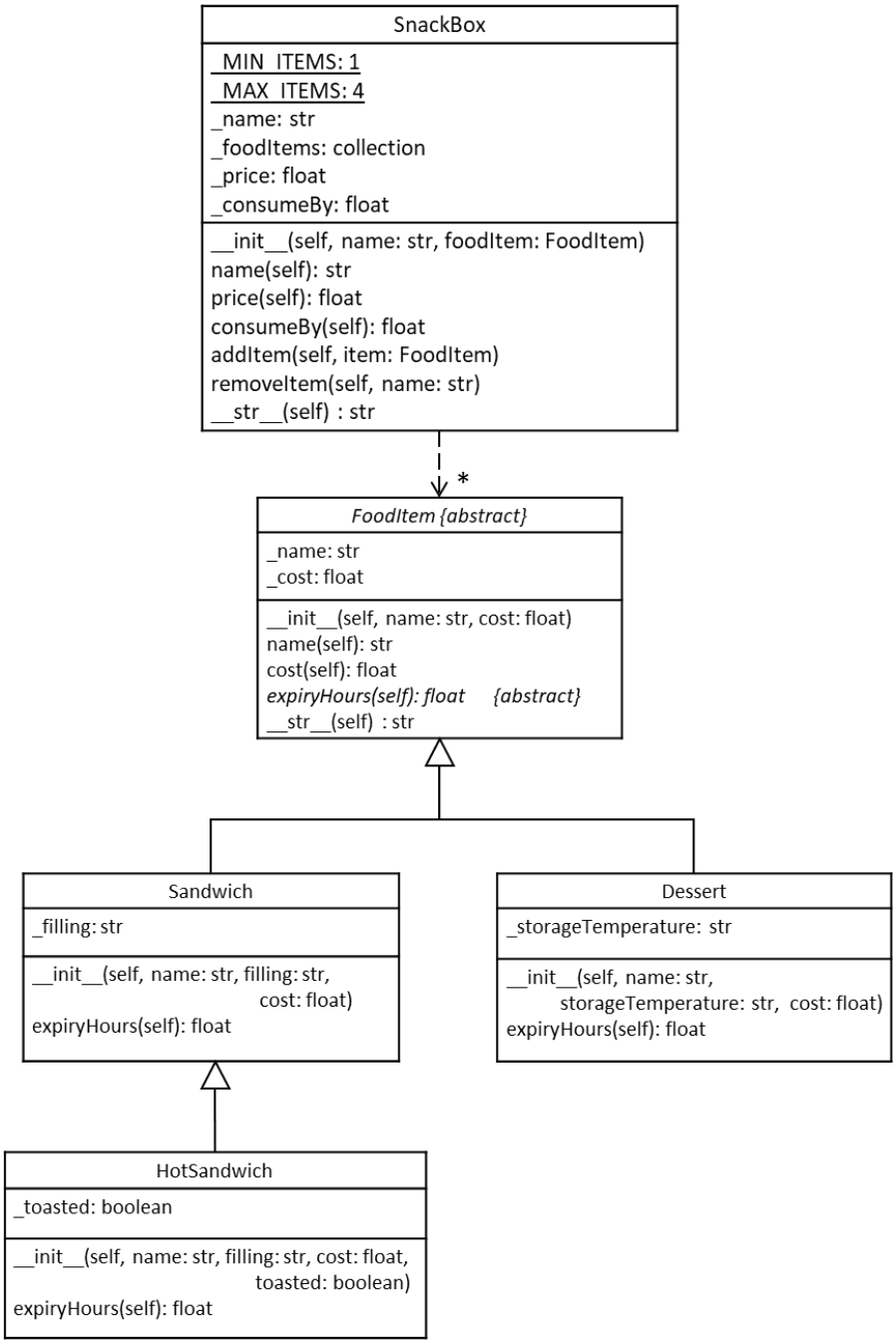

Figure Q3(a)

Question 3a

Write a Python class definition for SnackBoxException, which is a subclass of the Exception class. This class has no additional attribute or method.

(1 mark)

Question 3b

Implement the SnackBox class.

The SnackBox class represents a Snack box with selected FoodItem assembled by a customer. It has the following attributes:
Two class variables:

- _MIN_ITEMS and _MAX_ITEMS: representing the minimum and maximum number of FoodItem that the snack box can contain.

Four instance variables:
- _name represents the customer defined name for this snack box.
- _foodItems is a collection containing the customer selection of FoodItem objects for this snack box. It is possible that a customer can add the same FoodItem (e.g., 2 Toasted Cheese Sandwiches) into the snack box. You can choose to implement either Dictionary or List for this collection.
- _price represents the final price of the assembled snack box, which is the sum of all FoodItem's cost.
- _consumeBy is the lowest expiry hours of all FoodItem objects.

The constructor performs the following:
- Initialise _name with the given parameter name.
- Set _foodItems to an empty collection, then add the given parameter foodItem into the collection.
- _price and _consumeBy are set to the cost and expiryHours of the given parameter foodItem.

Getter methods for name, price and consumeBy. Use the property decorator.
addItem method adds the given parameter foodItem into the collection _foodItems.

- If this addition results in _foodItems having more than _MAX_ITEMS, raise SnackBoxException with appropriate message.
- Ensure _price and _consumeBy is re-computed after each successful addition.

removeItem method removes from _foodItems, the first FoodItem object that matches the parameter name.

- If there is no foodItem matching the given name, then method should raise SnackBoxException with appropriate message.
- If the number of foodItems after removal is less than _MIN_ITEM, do not allow the removal and raise SnackBoxException with appropriate message.
- Ensure _price and _consumeBy is re-computed after each successful removal.

The __str__ method returns a string representation of a SnackBox object, including the name of each FoodItem objects.

This is an example in the suggested format:

```
Name: Revive Snack      Price: $7.50     Consume in 2.0 hrs
      with Cold Cut Sandwich, Tiramisu
```

(15 marks)

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Question 3c

State at least **TWO (2)** reasons to justify your choice of collection for the implementation of SnackBox's _foodItems.

(5 marks)

Question 3d

Write Python statements to do the following:

Question 3d(i)
Create the appropriate FoodItem objects using the information in Table Q3(d)(i) below.

Dessert Name	Cost	Storage Temperature
Chendol	$3.50	4
Cheng Teng	$3.20	16
Ice Kachang	$3.50	4
Pulut Hitam	$3.50	38
Bubur Cha Cha	$4.20	38

Table Q3(d)(i)

(2 marks)

Question 3d(ii)
Create 2 snack boxes, sb1 ('His Desserts') and sb2 ('Her Dessert'), with Chendol and Cheng Teng as their first FoodItem respectively.

(1 mark)

Question 3d(iii)
Add these desserts to sb1 in this order: Cheng Teng, Ice Kachang, Pulut Hitam and Bubur Cha Cha. Ensure exceptions are properly handled.

(2 marks)

Question 3d(iv)
Remove Cheng Teng from sb2. Ensure exceptions are properly handled.

(2 marks)

Question 3d(v)
Based on your understanding, what will be the output from these python statements, if exceptions are handled properly in Q3(d)(iii) & Q3(d)(iv)?

```
print(sb1)
print(sb2)
```

(2 marks)

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Question 4

A prototype GUI is developed to allow students to enquire for their TMA and TOA results. The GUI allows student to enter their student Id, select TMA or TOA, and click "Enquire" to find out if they "Pass", "Fail" or the marking is still "In Progress". The initial design of the GUI is shown in Figure Q4(a).

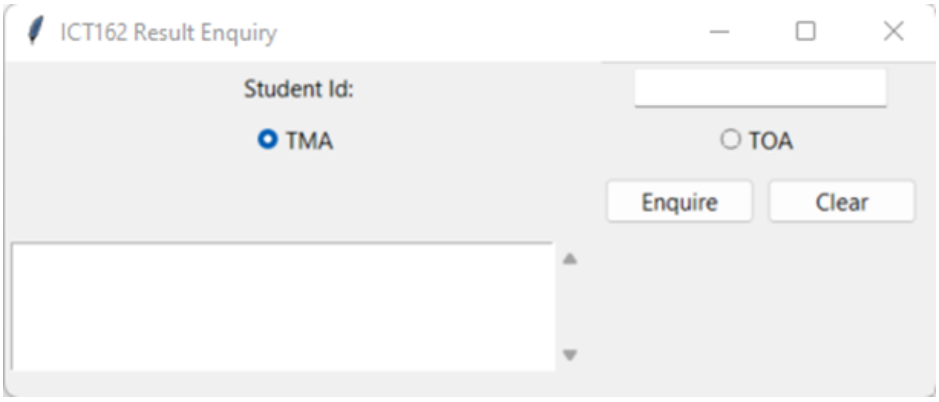

Figure Q4(a)

The following is the initial code for the GUI shown in Figure Q4(a).

```
import tkinter as tk
from tkinter import ttk, scrolledtext

class ResultGUI:
    def __init__(self):
        self._win = tk.Tk()
        self._win.geometry('380x300')
        self._win.title('ICT162 Result Enquiry')
        self.create_widgets()
        self._win.mainloop()

    def create_widgets(self):
        # create widgets
        self._studentID = tk.StringVar()
        self._lblId = ttk.Label(text='Student Id:')
        self._txtId = ttk.Entry(width=20, textvariable=self._studentID)
        self._rbtn = tk.IntVar()
        self._rbtnAm = ttk.Radiobutton(text='TMA', value = 0, variable=self._rbtn)
        self._rbtnPm = ttk.Radiobutton(text='TOA', value = 1, variable=self._rbtn)
        btnFrame = ttk.Frame(self._win)
        self._btnEnquire = ttk.Button(btnFrame, text='Enquire',
command=self.enquire)
        self._btnClear = ttk.Button(btnFrame, text='Clear')
        self._sclText = scrolledtext.ScrolledText(width=35, height=4)
        self._sclText.config(state=tk.DISABLED)
        self._txtId.focus()

        # Placing the widgets using Grid layout manager
        self._lblId.grid(row=0, column=0, padx=4, pady=4)
        self._txtId.grid(row=0, column=1, padx=4, pady=4)
        self._rbtnAm.grid(row=1, column=0, padx=4, pady=4)
        self._rbtnPm.grid(row=1, column=1, padx=4, pady=4)
        self._btnEnquire.grid(row=0, column=0, padx=4, pady=4)
        self._btnClear.grid(row=0, column=1, padx=4, pady=4)
        btnFrame.grid(row=3, column=1, padx=4, pady=4)
        self._sclText.grid(row=4, column=0, padx=4, pady=4)

def enquire(self):
    pass

ResultGUI()
```

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Question 4a

The user has commented that the scrolled text is not properly placed and the radio buttons for TMA and TOA should be align at the right as shown in Figure Q4(b).

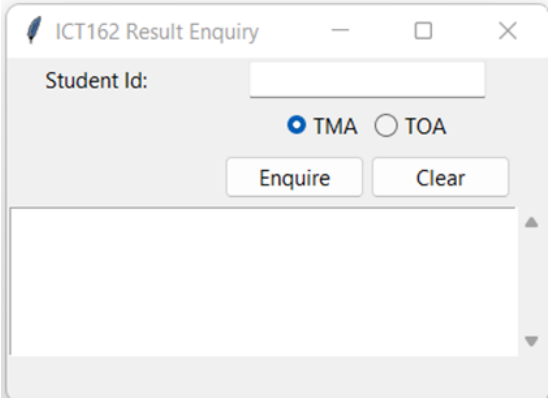


Figure Q4(b)

Based on the code given, write the necessary changes to correct the placement of the scrolled text and reflect the new position of the radio buttons.

(8 marks)

Question 4b

For simulation purpose, the following data structure is setup to contain the outcomes for TMA and TOA. In each assessment, 'Pass', 'Fail' and 'In Progress' are keys and the values are the list of student Ids.

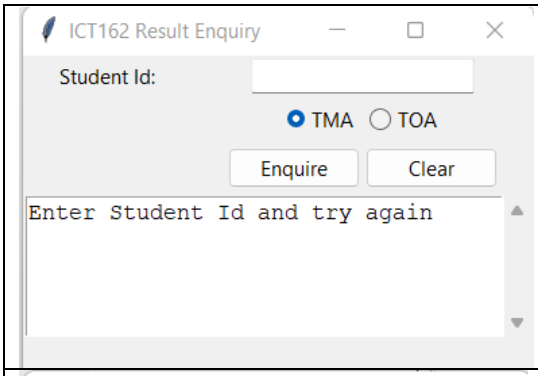
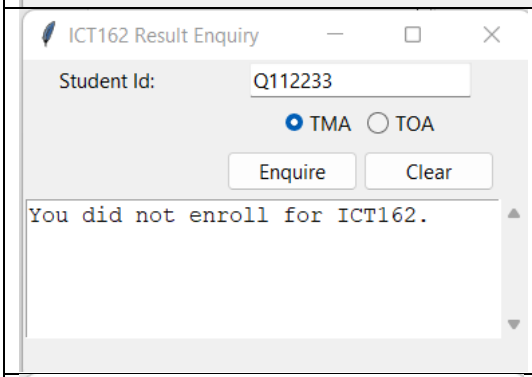
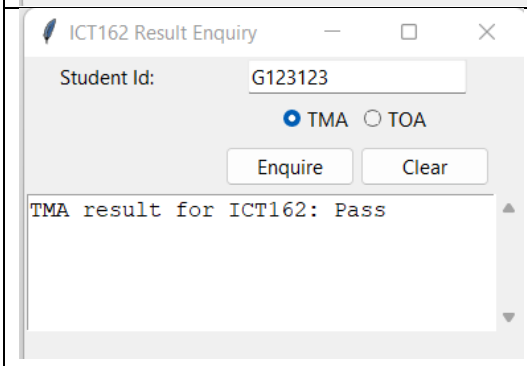
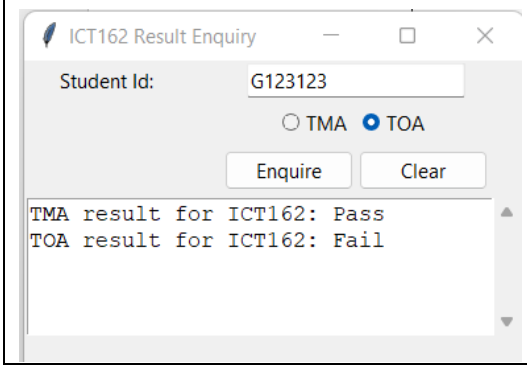
```
{ 'TMA': { 'Pass': ['G123123','E223344'],
           'Fail': ['E177890','M563746'],
           'In Progress': ['M998877','G432432'] },
  'TOA': { 'Pass': ['E223344'],
           'Fail': ['G123123'],
           'In Progress': ['E177890','M563746','M998877','G432432'] }
}
```

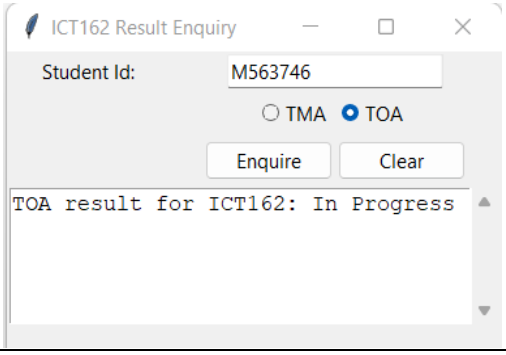
Indicate how this dictionary structure should be declared based on code given. Indicate the line number where the statement should be inserted.

(4 marks)

Question 4c

Write statements that will implement the event handler for Enquire button. Below are demonstrations of the ResultGUI:

	If Enquire button is clicked when Student Id is empty, display a message.
	If Student Id is not found in that assessment, display a message.
 	If Student Id is found in the selected assessment, display the outcome for this student.

Submit your codes for the enquire() function.

(13 marks)

----- END OF PAPER -----